

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
 Department of Computer Science and Engineering
ASSESSMENT TOOL 3 – TECHNICAL PRESENTATION

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	Assessment:	Assessment Tool 3 – Technical Presentation
Weightage:	10% of CIA	Total Marks:	100 (1 Question × 100 Marks)
CO1 Statement:	<i>Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)</i>		
Student Name:	<i>Sasikumar Megha</i>	Reg. No.:	<i>192471038</i>
Section / Batch:		Date:	

Instructions to Students

- Answer the given question through a technical presentation. Total Marks: 100.
- Clearly define the problem and explain the concept of algorithm growth functions.
- Present comparative analysis using appropriate representations such as graphs, tables, or charts.
- Use suitable tools (PowerPoint, visualization tools, or software) to support your explanation.
- Ensure technical accuracy, logical flow, and proper interpretation of results.
- Maintain clarity in communication, proper slide organization, and professional delivery.
- Time management, confidence, and audience engagement will be considered during evaluation.

Question Type	Focus Area	Questions	Marks
Technical Presentation	Analyze and compare growth functions using mathematical techniques and asymptotic concepts and with graphical and visual interpretation	Q1	100
GRAND TOTAL:			100 Marks

SECTION A – Technical Presentation

15

50

Code of Conduct

Sasikumar Megha (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: S. Megha

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Marks Evaluation:

Question No.	Technical Content Accuracy (30%)	Problem Identification (25%)	Use of Tools (20%)	Communication (15%)	Professionalism (10%)	Total Marks (100)
Q1	29	24	20	14	8	45
Overall Total (100)						

Faculty In-charge	Course Coordinator	Head of Department
Signature: <u>[Signature]</u>	Signature: _____	Signature: _____
Date: <u>20/11/26</u>	Date: _____	Date: _____

Q) A Symptotic Analysis of Disjoint Set Operations
 Analyze the efficiency of find and union operations in the disjoint set data structure using asymptotic notation. clearly define the problem and explain the role of parent links in operations. Use diagrams or small examples to illustrate how the structure evolves. Present the findings clearly and interpret the efficiency of this approach in graph algorithms.

A) Disjoint Set (union-find) maintain separate non-overlapping sets using trees. Each element has a parent link, and the root represents the entire set.

It support two operations:

* Find(x): Follow parent links to find the root of (x).

* union(x, y): finds the roots of x and y then joins their sets.

Example:

Initially: 1 2 3

After union (1, 2):



After union (1, 3):



Time Complexity

Method	find	union
Basic implementation	$O(n)$	$O(n)$
union by rank	$O(\log n)$	$O(\log n)$

Path compression + union by rank $O(\alpha(n))$

Here $\alpha(n)$ is the inverse Ackermann function, which grows very slowly. Therefore $O(\alpha(n))$ is considered nearly $O(1)$ in practice.

The space complexity is $O(n)$

Disjoint set is used in Kruskal's Minimum Spanning Tree algorithm.

Q 92)

Master Theorem Application for Balanced Quadratic Recurrence.

Consider the recurrence relation $T(n) = 4T(n/2) + n^2$. Analyse the problem using the Master Theorem and determine its time complexity. Clearly identify the parameters and compare $f(n)$ with $n^{\log_b a}$. Justify the selected case and explain the reasoning step by step. Use recursion tree support if needed and present the final result with interpretation.

Given:

$$T(n) = 4T(n/2) + n^2$$

Master Theorem form

$$T(n) = aT(n/b) + f(n)$$

Here

$$a = 4$$

$$b = 2$$

$$f(n) = n^2$$

Calculate

$$n \log_b a = n \log_2 4 = n^2$$

Since

$$f(n) = O(n^{\log_b a}) = O(n^2)$$

Master Theorem (Case 2) applies.

$$T(n) = O(n^2 \log n)$$

Therefore :

$$T(n) = O(n^2 \log n)$$

Each recursion tree level cost n^2 , and there are $\log_2 n$ levels. Hence the total complexity is $O(n^2 \log n)$.

Q.3) Substitution Method for Constant Increment Recurrence.

Analyse the recurrence $T(n) = T(n-1) + 5$ using the Substitution method. clearly define the recurrence and apply repeated expansion until a summation or direct pattern is obtained. Justify each step mathematically and simplify the expression fully. Use structured steps or a small table if required for clarity. Present the final asymptotic complexity and interpret the efficiency of the algorithm.

Given

$$T(n) = T(n-1) + 5$$

Repeated substitution:

$$T(n) = T(n-2) + 5 + 5$$

$$= T(n-3) + 15$$

$$= T(n-k) + 5k$$

Let $k = n - 1$ Then :

$$T(n) = T(1) + 5(n-1)$$

If $T(1) = c$:

$$T(n) = c + 5n - 5$$

Ignoring constants

$$T(n) = O(n)$$

Therefore, the running time grows linearly with the input size.

Q 94) Comparison of exponential and polynomial Growth.
Compare algorithm with complexities $O(n^2)$, $O(n^3)$, and $O(2^n)$ in terms of feasibility for large inputs. Clearly define the problem of choosing an algorithm for computationally intensive tasks. Use graphs or tables to illustrate how the growth rates diverge as n increases. Explain why exponential complexity. Present conclusion on algorithms selection for scalable applications.

Consider three algorithms :

* $O(n^2)$: Quadratic

* $O(n^3)$: Cubic

* $O(2^n)$: Exponential

Growth comparison

n	n^2	n^3	2^n
5	25	125	32
10	100	1000	1,024
20	400	8000	1,048,576
30	900	27,000	1,073,741,824

for small inputs, all three may be usable. However 2^n increases extremely quickly and becomes impractical even for moderate input sizes.

Therefore

$$O(n^2) < O(n^3) < O(2^n)$$

Q 95) Recursion Tree Analysis

Analyse the recurrence $T(n) = 3T(n/3) + n$ using the recursion tree method. Clearly construct the recursion tree and determine the cost at each level. Sum the costs across all level to obtain total complexity. Use diagrams or structured calculations to support the analysis. Interpret the result and compare it with the Master Theorem solution.

Given

$$T(n) = 3T(n/3) + n$$

At the root level, the cost is n

At the next level $n/3$: $3 \times \frac{n}{3} = n$

Level 0: n

Level 1: $n/3$ $n/3$ $n/3$

Total cost = n

Level 2: 9 subproblems of size $n/9$

Total cost = n

The tree has: $\log_3 n$

level therefore:

$$T(n) = n \times \log_3 n$$

Hence: $T(n) = O(n \log n)$

$$a = 3, b = 3, f(n) = n$$

$$n \log_3 3 = n$$

Since $f(n) = (-)(n)$, case 2 applies and gives the same result:

$$O(n \log n)$$